

A Mini Project Report
On
DETECTION OF DROWSINESS AND ACCIDENT PREVENTION.

Submitted in partial fulfillment of the requirement for the award of degree

Of

BACHELOR OF TECHNOLOGY
in
ELECTRONICS AND COMMUNICATION ENGINEERING.

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VIGNAN'S LARA
INSTITUTE OF TECHNOLOGY & SCIENCE
(AUTONOMOUS)

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Vadlamudi - 522 213, Guntur District

2022-2026



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CERTIFICATE

This is to certify that, this dissertation entitled “ **DETECTION OF DROWSINESS AND ACCIDENT PREVENTION.**” is a bonafide work done by **K.RESHMA(22FE1A0472), P.JAYASREE(22FE1A0454),K.KRISHNAVAMSI(22FE1A0455),K.SHYAMSUNDAR(22FE1A0457)**, under my guidance and submitted in partial fulfillment of the requirement for the award of the degree of **Bachelor of Technology in Electronics and Communication Engineering** by **Jawaharlal Nehru Technological University, Kakinada.**

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DECLARATION

We hereby declare that the project entitled “**DETECTION OF DROWSINESS AND ACCIDENT PREVENTION**” has been undertaken by us and this work has been submitted to VIGNAN’S LARA INSTITUTE OF TECHNOLOGY AND SCIENCE affiliated to **JNTUK, Kakinada** in the partial fulfillment of the requirements for the award of **BACHELOR OF TECHNOLOGY (B. Tech)** in the department of **Electronics and Communication Engineering**, is the result of the work done by us under the guidance of **Ms.SANDHYARANI**, Assistant Professor of ECE department.

The work is original and has not been submitted for any Degree/Diploma of this or any other university.

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ABSTRACT

This mini-project focuses on the detection of drowsiness as a critical factor in accident prevention, particularly in the context of road safety. With the increasing prevalence of drowsy driving, it is essential to develop effective monitoring systems that can identify signs of fatigue in real-time. The project utilizes a combination of physiological and behavioral indicators, such as eye-tracking, facial recognition, and heart rate variability, to assess driver alertness. Through the integration of machine learning algorithms, the system aims to provide timely alerts to the driver when signs of drowsiness are detected. A prototype was developed and tested in simulated driving scenarios, demonstrating the potential to reduce the risk of accidents caused by drowsiness. The findings underscore the importance of proactive measures in enhancing road safety and suggest further research directions for refining detection technologies. This project contributes to the broader field of intelligent transportation systems, aiming to create safer driving environments and promote public awareness of the dangers of drowsy driving.

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1. INTRODUCTION

Drowsiness during driving is a significant safety concern, often leading to impaired judgment, slower reaction times, and reduced awareness of the driving environment. The Anti-Sleep Alarm project is designed to address the critical issue of drowsiness, particularly in contexts like driving, working long shifts, or studying. With sleep-related accidents and reduced productivity being significant concerns, this project aims to create a wearable device that detects signs of fatigue through physiological indicators such as heart rate variability, blink patterns, and posture. When the device senses that the wearer is becoming drowsy, it triggers a series of alerts, such as vibrations, sounds, or visual signals, to encourage them to stay alert. By leveraging technology to promote awareness and safety, the Anti-Sleep Alarm project seeks to enhance well-being and prevent accidents, ultimately fostering a more alert and productive society.

1.1 Key Components

- Drowsiness Detection:

The project utilizes a combination of wearable sensors, cameras, and machine learning algorithms to monitor physiological indicators of drowsiness. Tracking blink frequency and duration to detect fatigue. Analyzing heart patterns to assess stress and alertness.

- Data Analysis:

The collected data is processed in real-time to identify patterns associated with drowsiness. Machine learning models are trained to recognize the early signs of fatigue based on historical data.

- Accident Prevention:

The primary goal of the project is to prevent accidents caused by drowsiness. By providing timely alerts, the system encourages individuals to take breaks, engage in activities to re-energize, or, in critical situations, safely pull over.

The Driver Anti-Sleep Alarm project is a safety-focused initiative aimed at reducing the risk of drowsiness-related accidents on the road. This project involves developing a system that monitors driver alertness through a combination of sensors, such as eye-tracking cameras and accelerometers, which detect signs of fatigue like prolonged eye closure and head nodding. When drowsiness is detected, the system activates auditory and visual alerts to prompt the driver to take a break or refocus their attention. By integrating this technology into vehicles, the project aims to enhance road safety, reduce accidents, and promote responsible driving habits, ultimately contributing to safer travel experiences for all road users.

Drowsiness-related accidents occur when drivers fall asleep or become significantly less alert while operating a vehicle. Fatigue impairs cognitive functions such as reaction time, decision-making, and attention, increasing the likelihood of crashes. Common signs of drowsiness include frequent yawning, heavy eyelids, and drifting out of lanes. Research shows that drowsy driving can be as dangerous as driving under the influence of alcohol, leading to severe accidents that can result in injuries or fatalities. Long trips, irregular sleep patterns, and certain medications are common contributors to driver fatigue, highlighting the need for awareness and preventive measures to ensure road safety.

2.OBJECTIVE

The main objective of the project focused on the detection of drowsiness and accident prevention is to develop a reliable and efficient system that identifies signs of driver fatigue in real-time, thereby reducing the risk of accidents caused by drowsy driving. Here are the key components of this objective:

- Real-Time Drowsiness Detection
 - Implementing advanced monitoring techniques, such as computer vision and machine learning, to accurately assess driver alertness based on physiological indicators (e.g., eye movement, blink rate) and behavioral patterns (e.g., head position).
- Timely Alerts and Interventions
 - Designing a system that provides immediate feedback to the driver when signs of drowsiness are detected. This could include auditory alerts, visual cues, or vibrations to prompt the driver to take a break or refocus.
- Data Analysis and Interpretation
 - Utilizing data analytics to identify trends and patterns associated with drowsy driving, which can inform further improvements in detection methods and preventive measures.
- Integration with Vehicle Systems
 - Ensuring compatibility with existing vehicle technologies to enhance overall safety features. This could involve integrating the drowsiness detection system with advanced driver- assistance systems (ADAS).
- Awareness and Education
 - Raising awareness about the dangers of drowsy driving and promoting the use of the detection system among drivers, thereby fostering a culture of safety on the road.
- Long-Term Safety Improvement
 - Ultimately, the project aims to contribute to a significant reduction in drowsy driving incidents, thereby enhancing overall road safety and decreasing the incidence of related accidents and fatalities.

By achieving these objectives, the project seeks to create a proactive approach to drowsy driving, leveraging technology to safeguard drivers and promote safer roads for everyone.

3. LITERATURE SURVEY:

The issue of drowsy driving and its contribution to road accidents has garnered significant attention in recent years. This literature survey reviews various approaches and technologies developed for the detection of drowsiness and strategies for accident prevention.

- Physiological Indicators of Drowsiness
 - Eye Movement and Blink Rate: Several studies, such as those by Wang et al. (2018), utilize computer vision techniques to monitor eye movements and blink patterns. Changes in blink frequency and duration can serve as key indicators of fatigue.
 - Heart Rate Variability: Research by Veen et al. (2020) shows that heart rate variability can provide insights into driver fatigue. Lower variability is often associated with increased drowsiness, suggesting the potential for wearable devices to monitor this metric.
- Machine Learning and Computer Vision
 - Many studies leverage machine learning algorithms to improve drowsiness detection systems. For instance, Saha et al. (2022) implemented convolutional neural networks (CNNs) to analyze images of drivers and detect signs of fatigue with high precision.
 - A hybrid approach combining multiple indicators (physiological and behavioral) has been proposed by Liu et al. (2023), demonstrating improved accuracy over single-modality systems.
- Wearable Technology
 - Research has shown the effectiveness of wearable devices, such as smartwatches, to monitor physiological signals (Kim & Lee, 2020). These devices can alert drivers when drowsiness is detected, promoting timely intervention.

4. METHODOLOGY

4.1 BLOCK DIAGRAM

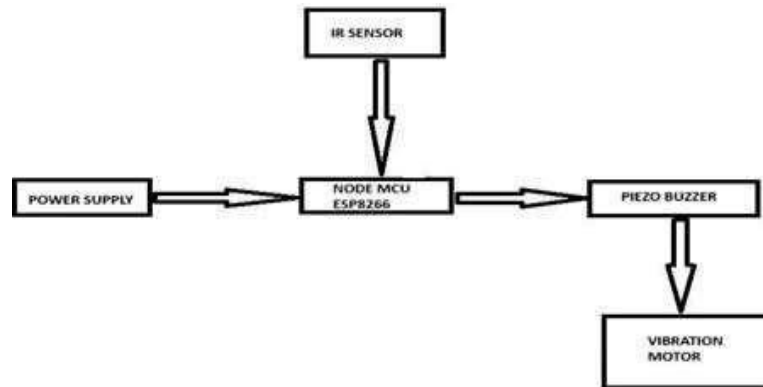


Fig – 1: Block Diagram of Anti-Sleep Alarm

4.2 .NODE MCU

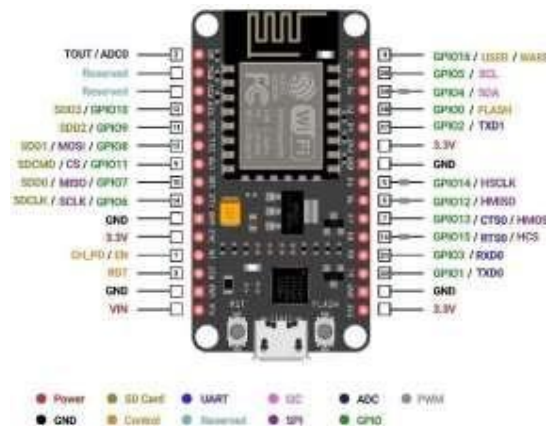


Fig.2 Node MCU

Node MCU Development board is featured with Wi-Fi capability, analog pin, digital pins, and serial communication protocols. After setting up ESP8266 with Node-MCU firmware Hardware Features:

- **ESP8266 Module:** The core of NodeMCU is the ESP8266 microcontroller, which integrates Wi-Fi capabilities, making it ideal for IoT projects.
- **GPIO Pins:** It provides multiple General Purpose Input/Output (GPIO) pins that can be used to connect sensors, actuators, and other components.
- **Analog Input:** NodeMCU features an ADC (Analog-to-Digital Converter) pin that can read analog signals.
- **USB Interface:** A built-in USB interface simplifies programming and power supply.

The NODE MCU is equipped with 30 male I/O headers, in a DIP-30-like configuration, which can be programmed using the Arduino Software Integrated Development Environment (IDE), which is common to all Arduino boards and running both online and offline.

4.3 RELAY:



Fig.3 Relay

A relay is an electromechanical switch that uses an electromagnetic coil to control the opening and closing of one or more sets of contacts. It allows a low-power signal to control a higher-power circuit, making it a crucial component in various applications, from automotive systems to industrial automation.

Specifications:

Coil Voltage: The voltage required to energize the coil, typically available in various ratings (e.g., 5V, 12V, 24V, 120V AC/DC).

Contact Rating: Specifies the maximum load the relay contacts can handle, usually expressed in volts (V) and amperes (A). This includes:

- **DC Rating:** For direct current applications.

4.4 IR SENSOR:

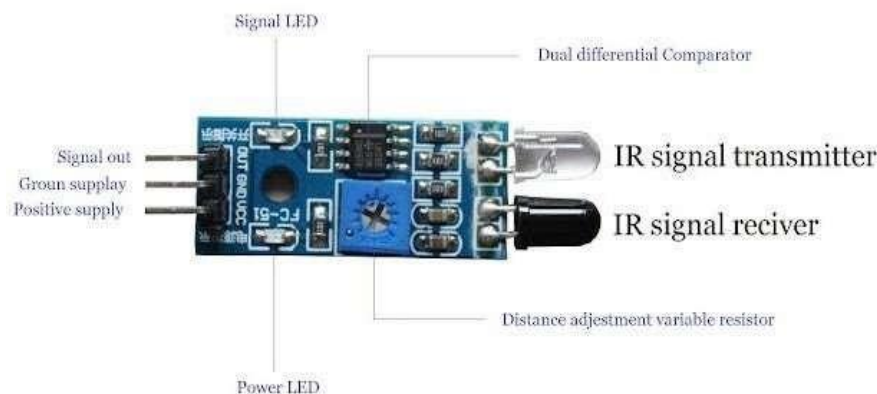


Fig.4 IR Sensor

IR Sensor is an electronic device that emits the light in order to sense some object of the surroundings. An IR Sensor can measure the heat of an object and detects the motion. Radiations which are invisible to eyes, can be detected by this sensor. The emitter is simply an IR LED and the detector is IR Photo diode.

The emitter is simply an IR LED and the detector is simply an IR Photodiode. Photodiode is sensitive to IR light of the same wavelength which is emitted by the IR LED. When IR light falls on the photodiode, the resistances and the output voltages will change in proportion to the magnitude of the IR light received. There are five basic elements used in a typical Infrared detection system: an Infrared source, Transmission medium, Optical component, Infrared Detectors or Receivers and Signal Processing.

4.5 BUZZER:

The Piezo Buzzer produces sound based on reverse of the Piezoelectric effect. The generation of pressure variation or strain by the application of electric potential across a piezoelectric material is the underlying principle. These buzzers can be used alert a user of an event corresponding to a switching action, Counter signal or Sensor input. They are also used in alarm circuits. The buzzer produces a same noisy sound irrespective of the voltage variation applied to it. It consists of Piezo crystal between two conductors. When a potential is applied across these crystals, they push on one conductor and pull on the other. This, push and pull action, results in a sound wave. Most buzzers produce sound in the range of 2 to 4 KHz. The Red lead is connected to the input and the Black lead is connected to ground.



Fig.5 Buzzer.

4.6 LED:



Fig.6 LED

A semiconductor light source called a light-emitting diode (LED) produces light when current passes through it. LEDs have various benefits, such as lower power consumption, a longer lifespan, increased physical resilience, smaller size, and quicker switching.

Specifications of an LED:

- Forward Voltage (V_f): The voltage drop across the LED when it is forward-biased, typically ranging from 1.8V to 3.6V, depending on the color and type.
- Forward Current (I_f): The current that the LED is designed to operate with, usually measured in milliamperes (mA). Common values are 20 mA for standard LEDs.
- Luminous Intensity: Measured in millicandelas (mcd), it indicates how bright the LED appears at a specific current and angle.
- Wavelength (**Color**): The wavelength of light emitted by the LED, which determines its color (e.g., red, green, blue).
- Power Rating: The maximum power the LED can handle, typically expressed in milliwatts (mW).
- Lifespan: LEDs have a long lifespan, often exceeding 25,000 hours, depending on operating conditions.

In summary, LEDs are versatile components in electric circuits, providing efficient and reliable light output for a wide range of applications. Their ability to be easily integrated into various designs makes them a preferred choice in modern electronics.

4.7 DC Motor:



Fig.7 DC Motor

A DC motor is an electromechanical device that converts direct current electrical energy into mechanical energy. It is widely used in various applications, from small toys to large industrial machinery, due to its simplicity, efficiency, and ease of control. The nominal voltage at which the motor is designed to operate (e.g., 6V, 12V, 24V). The maximum current the motor can draw under load, typically expressed in amperes (A). The power output of the motor, usually measured in watts (W) or horsepower (HP). The rotational speed of the motor, measured in revolutions per minute (RPM). The ratio of mechanical power output to electrical power input, expressed as a percentage. DC motors are fundamental components in electric circuits, providing efficient and controllable mechanical power for a wide range of applications. Their versatility and ease of integration make them a popular choice in both consumer and industrial products.

7. Eye Glasses/Spectacles:

The eye blink system comes with an IR sensor mounted on glasses which wear like regular glasses, shown in the picture below. Eye blink Sensor is a relatively simple Sensor used to detect eye blinks. It uses a simple infrared Sensor to detect if the person's eye is closed and the corresponding data and that can be used and it can be used and it can be received can further be processed by any logic as required for the application. This eye glasses should be wear by the driver when he was driving.

5.CIRCUIT DIAGRAMS

- Circuit Model:

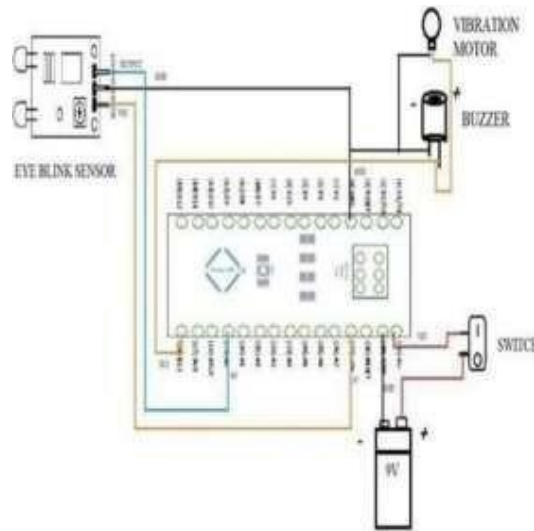


Fig .8 Circuit diagram-connections

- Features:
 - Microcontroller: Ten silica 32-bit RISC CPU Xtensa LX106
 - Operating Voltage: 3.3V
 - Input Voltage: 7-12V
 - Digital I/O Pins (DIO): 16
 - Analog Input Pins (ADC): 1

6.PROCEDURE

- Note the requirements:

Detect early signs of sleep or drowsiness (e.g., head tilting, prolonged eye closure, lack of motion) and sound an alarm. Drivers, machine operators, or workers in environments where staying awake is crucial.

- Selection of the components:

Detects prolonged eye closure, which may indicate drowsiness. Detects head tilting, which may suggest drowsiness. Changes in heart rate can also indicate drowsiness. For facial recognition and blink rate detection. Arduino, Raspberry Pi, or ESP32 for processing the input from sensors and triggering the alarm .Battery or USB power source for portability. To show the sensor readings.

- Circuit Design:

Connect the chosen sensor (eye blink, accelerometer, or other) to the microcontroller. Connect to an analog or digital input pin. Use an I2C or SPI interface to read tilt/movement data. Connect the alarm (buzzer or vibration motor) to a digital output pin of the microcontroller. Ensure the circuit is powered properly (via batteries or a DC source).

Pin	Function
RST	Reset the Module
ADC0	ADC Pin with 10-bit resolution
EN	Chip Enable Pin (active HIGH)
GPIO16	GPIO16 pin (wake pin from deep sleep mode)
GPIO14	GPIO14 pin (HSPI_CLK)
GPIO12	GPIO12 pin (HSPI_MISO)
GPIO13	GPIO13 pin (HSPI_MOSI)
VCC	3.3V Power Supply (max 3.6V)
SDCMD	SDIO CMD (GPIO11)
SD00	SDIO Data 0 (GPIO7)
SD02	SDIO Data 2 (GPIO9)
SD03	SDIO Data 3 (GPIO10)
SD01	SDIO Data 1 (GPIO8)
SDCLK	SDIO CLK (GPIO6)
GND	Ground Pin
GPIO15	GPIO15 pin (HSPI_CS)
GPIO2	GPIO2 pin (TXD1)
Flash	Flash Pin (GPIO0)
GPIO4	GPIO4 pin (SDA – software I2C)
GPIO5	GPIO5 pin (SCL – software I2C)
RXD0	UART0 RXD pin (GPIO3)
TXD0	UART0 TXD (GPIO1)

Fig.9 Function of Node MCU ESP2866

7.WORKING

The anti-sleep alarm project is designed to detect signs of drowsiness and alert the user to stay awake. Here's a breakdown of how it works:

1. Input:

The input to the circuit is given through the power supply and to the DC motor by the 9V supply Battery. The node mcu gets activated when the input is given. After the input is given the whole circuit is under on condition .

2. Intermediate Process:

The rotation of the wheel which is connected to the DC motor represents the movement of the vehicle. The relay will be under off condition until it detects any movement. The person/driver will be wearing those sensor connected glasses or spectacles, when the driver feels drowsy he tends to close his eyes, so at that moment when the closed eyelid comes in contact with the sensor it recognizes and warns an alarm buzzer and the led glows indicating the danger.

3. Output:

The output of this project can be seen in the buzzer of the alarm which rings at the moment where The sensor recognizes the movement of the eye lid and the relay comes into active state and controls the operation of remaining components .The DC motor becomes inactive and the wheel slows down in practical case it represents the vehicle.

4. Conclusion:

Hence, the Anti-Sleep Alarm project helps us to detect the drowsiness of the driver and prevents the occurrence of the accidents. These sensor based glasses are mostly useful for the people who are driving at night times, such as lorry drivers, bus drivers etc.The percentage of the accidents that occur during nights and due to drowsiness can be controlled by this project.

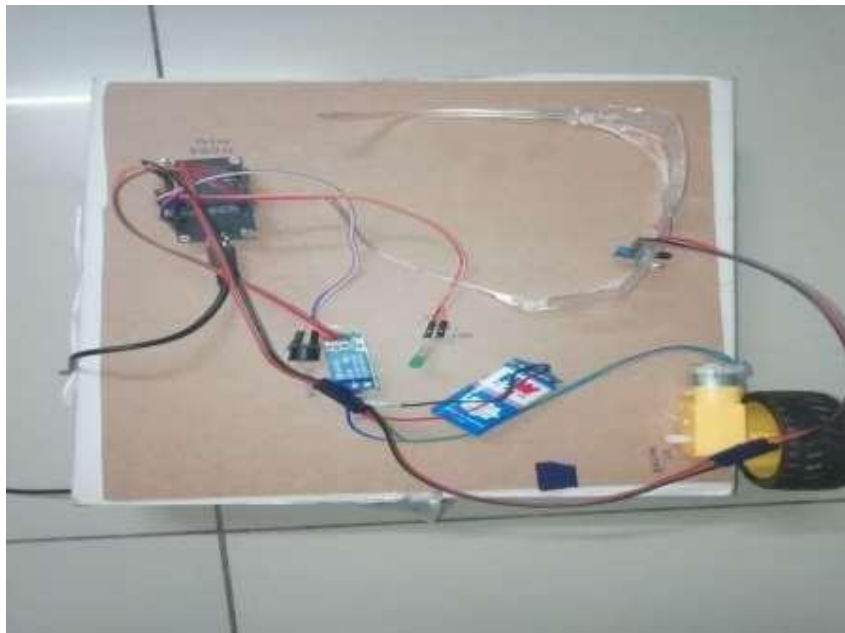


Fig.10Anti -Sleep Alarm.

8.CODE

```
#include <LiquidCrystal.h>

LiquidCrystal lcd(0x27, 16,2);

const int BUZZ=D5;

const int LED=D6;

const int RELAY=D7;

const int EYESENSOR=D0;

int Data;

void setup(){

  lcd.begin();

  lcd.backlight();

  pinMode(RELAY, INPUT);

  pinMode(BUZZ, INPUT);

  pinMode(LED, INPUT);

  pinMode(EYESENSOR, OUTPUT);

  Serial.begin(9600);

}

void loop(){

  Data=digitalRead(EYESENSOR);

  Serial.print(Data);

  if(Data==0){

    digitalWrite(LED, 0);

    digitalWrite(RELAY, 0);

    digitalWrite(BUZZ, 1);

    lcd.clear();

    lcd.setCursor(0,0);

    lcd.print("DANGER");

    delay(1000);

  }

}
```

```
else{  
    digitalWrite(LED, 1);  
    digitalWrite(RELAY, 1);  
    digitalWrite(BUZZ, 0);  
    lcd.clear();  
    lcd.setCursor(0,0);  
    lcd.print("NORMAL");  
    delay(1000);  
}  
}
```

8.CONCLUSION

8.1 Future Scope:

- Advanced Detection Algorithms: Future iterations could incorporate machine learning to improve the accuracy of fatigue detection through real-time data analysis from various sensors.
- Wearable Technology: Development of compact, wearable devices that monitor physiological signs of fatigue, making the system more accessible and user-friendly.
- Integration with Smart Devices: Linking the alarm system with smartphones and smart vehicles for enhanced alerts and feedback, allowing for more immediate responses to drowsiness.
- Customization and User Settings: Allowing users to customize alert thresholds and types of notifications, accommodating different environments and personal preferences.
- Research and Data Analysis: Utilizing collected data for broader research on sleep patterns and fatigue-related incidents, which can contribute to public safety campaigns.



Fig.11 Mini Project

8.2 Applications:

Automotive Industry: Implementation in cars and trucks to monitor driver alertness, reducing accidents caused by drowsiness.

Aviation: Use in cockpits to ensure pilots remain alert, especially during long flights or overnight shifts.

Healthcare: Assisting medical professionals during long shifts to maintain alertness and reduce medical errors.

Industrial Settings: Application in factories and warehouses to monitor workers operating heavy machinery, enhancing workplace safety.

Education: Utilizing in student environments to promote alertness during long study sessions or lectures.